Overlap-Correct Collar Charging and the Coherence–Charge Trade-off

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Abstract

We derive the multi-link free-energy consequence of continuous collar failure with overlap multiplicity retained. Combining it with connected-witness entropy exposes a quantitative incompatibility: inside the current nonlinear window, the coherence radius small enough for the coherent conditional bound is smaller than the radius needed to suppress incoherent-collar contours, even under ideal overlap one. The deficit yields explicit repair targets.

1 Overlap-correct deterministic charge

Let $W_1, \dots, W_k$ be selected collar-failure witnesses. Stage 16 implies that each witness has incident squared-curvature energy at least $\eta^2/2$. Assume each plaquette occurs in at most ν selected incident families.

Theorem 1 (Multi-witness charge)**.** *The global squared-curvature energy and its heat-kernel Gibbs exponent satisfy*

$$E_{\mathrm{global}} \geq \frac{k\eta^2}{2\nu}, \qquad \frac{E_{\mathrm{global}}}{2t} \geq \frac{k\eta^2}{4\nu t}.$$

Proof. Summing the k local inequalities gives $k\eta^2/2$. Every global plaquette-energy term is counted at most ν times, so the sum of local energies is at most νE_{global}. Division by ν and then $2t$ gives the result. □

The factor ν is necessary. Repeating one identical witness k times has $\nu = k$ and produces only one local charge; a naive disjoint sum overstates the exponent by exactly k.

2 Connected entropy

Let D be the witness adjacency degree and let $e^{\alpha k}$ be the retained state weight. With

$$a_\nu(\eta) = \frac{\eta^2}{4\nu t}, \qquad \theta = e^{\alpha} D^2 e^{-a_\nu(\eta)},$$

the standard rooted connected-set count gives, for minimum size k_0,

$$\sum_{k \geq k_0} e^{\alpha k} D^{2(k-1)} e^{-k a_\nu(\eta)} \leq \frac{e^{\alpha k_0} e^{-k_0 a_\nu(\eta)} D^{2(k_0-1)}}{1-\theta}$$

whenever $\theta < 1$.

Corollary 1 (Entropy threshold)**.** *Strict convergence requires*

$$\eta > \eta_{\mathrm{ent}}(\nu) := \sqrt{4\nu t\big(\alpha + 2\log D\big)}.$$

The executable module also solves the stronger threshold at which the negative logarithm of the rooted sum exceeds 6.71292101139*.*

3 The common-parameter test

Freeze $t = 3 \cdot 10^{-6}$, $D = 18$, $\alpha = 0.1$, and the maximum sharp local radius 0.011453785959974176. At twice that radius, compute the largest η_{coh} whose optimized coherent conditional bound fits the remaining $K + R$ budget. Separately compute η_{FE} required by the rooted incoherent contour with $k_0 = 2$.

Proposition 1 (Quantified trade-off obstruction)**.** *The accompanying interval-certified calibration gives*

$$\eta_{\mathrm{coh}} < \eta_{\mathrm{FE}}$$

*already for ideal overlap multiplicity $\nu = 1$. Therefore the present coherent estimate, deterministic collar charge, and degree-*18 *connected-set count have no common coherence parameter inside the certified nonlinear window. Any $\nu > 1$ increases the deficit.*

This proposition is a no-go for the current proof mechanism, not for Yang–Mills theory. The calibration translates the deficit into two positive targets: reduce the effective connected branching degree below the reported ceiling, or amplify the deterministic exponent by the reported factor. A sector-aware polymer can also change the counting law rather than either local constant.

Remark 1 (Claim boundary)**.** *The contour theorem is conditional on a witness selection with declared overlap and on a normalized partition comparison. The programme has not yet constructed this selection uniformly over adaptive histories, boundary states, volumes, orbit charts, and topological sectors. Hence neither physical basin entry nor a mass gap follows.*